Tioluwani Akinloye

Dual Citizenship: Canada and Nigeria | **Native English Speaker**

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Education

Simon Fraser University - Master of Visual Computing, Expected 2028' Burnaby, BC
- *Relevant Coursework:* Machine Learning, Distributed Cloud Systems

University of Alberta - BSc Honors Computing Science, 2025' Edmonton, AB
- *Relevant Coursework:* Visual Recognition, AI Search & Planning, Program Analysis, Models of Computation

Experience

Simon Fraser University - Research Assistant Sep 2026 - Present
- Supporting Project 15004: Experimental Study with Virtual Humanoid/Robot Model Development.
- Developing simple virtual models, including joint-link systems, for humanoid and robot applications.
- Exploring open-source software design and development environments through applied tutorials.
- Designing and evaluating user interaction methods with the developed virtual models.

ScaleupMap - Full Stack Developer Feb 2025 - May 2026
- Built full-stack features using React, TypeScript, Go, and PostgreSQL for a platform capable of serving up to 1,000+ users.
- Engineered multiple Python-based external AI services leveraging Google Gemini, with APIs deployed on GCP.
- Designed and optimized 20+ relational database schemas using DBeaver, supporting scalable data workflows.
- Implemented multiple unit, API, and E2E tests; assessed API call efficiency and reduced redundant calls by up to 25%

Knead Technologies - Student Developer Jan 2024 - April 2024
- Designed an intuitive dashboard and implemented various features, including on-demand and routine donation tracking, using ReactJS (Typescript), PHP (Laravel), Apache, and MySQL, all running on Docker, capable of serving 200+ active users.
- Developed the closed marketplace page and designed UI/UX components to streamline user transactions and feature key metrics.
- Conducted E2E testing of the application & APIs using Cypress, & unit testing with PHPUnit, reducing production bugs by ∼20%

Projects

Battlezone - OpenGL 3D Graphics Project
- Recreated the classic Atari Battlezone game with custom OpenGL-powered 3D graphics rendering
- Implemented real-time 3D rendering pipeline including camera transforms, lighting, and collision detection
- Designed custom AI behaviours for enemy tanks, enabling dynamic pathfinding and combat responses
- Built in Python using Pygame for graphics rendering, with a focus on efficient implementation and optimization.

SmartVision - Visual Recognition App
- Developed a web application that allows users to train and interact with machine learning models for numerical recognition.
- Implemented a canvas feature enabling users to draw numbers and identify them using logistic regression (92% accuracy), feedforward neural network (96% accuracy), and convolutional neural network (98% accuracy), each achieving optimized training times.
- Built using Vue.js for frontend, Flask for the backend and PyTorch for implementing and training the machine learning models.

StudyAI - Video and Audio Learning Tool
- Developed a full-stack learning tool that provides an AI chatbot tailored to the content of whatever video you upload.
- Built using React.js for frontend, FastAPI for backend, and Firebase Realtime Database for the database, allowing users to upload videos or youtube links and generate transcripts, summaries and related content.
- Integrated openai's Whisper model for transcription, supporting video/audio files up to 25 million bytes (∼30 mins) with 90% accuracy.
- Implemented an AI chatbot using openai's API, that answers video-related questions, generates summaries, highlights key points, and creates quizzes with ∼95% accuracy in comprehension-based responses.

Skills

Languages | C, R, C++, PHP, Java, Python, Golang, Racket, Java/TypeScript
Technologies | Git, Docker, OpenGL, OpenCV, Cypress, Postman, PyTorch
Frameworks | Flask, Vue.js, Node.js, Laravel, FastAPI, React.js, Express.js
Concepts | OOP, TDD, REST, Agile, Scrum, UI/UX, MERN, MEVN, CI/CD

Additional Information

Awards: Academic Honors (2020), Distinguished Service Award (2020)
Certifications: International Baccalaureate Diploma (2020)
Achievements: Faculty of Science Dean's List (2023)